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Bug Report: Collision Not Detected When Spawning Actor with 'spawn' Function at Ego Vehicle's Location #6653

Open

GabrielZZZ opened this issue 8 minutes ago · 0 comments

GabrielZZZ commented 8 minutes ago • edited

CARLA Version: latest

Platform/OS: Ubuntu 20.04

Problem Description:

I have encountered a bug related to the `spawn` function in CARLA. When attempting to spawn an actor at a specific location (actor spawn location: `Location(x=107.999817, y=132.417389, z=1.000000)`, `Rotation(pitch=0.000000, yaw=49.000000, roll=0.000000)`), it should have resulted in a collision with the ego vehicle (`Location(x=107.999817, y=133, z=0.5)`). According to the expected behavior, the actor should not have been spawned at this location to avoid the collision. However, in the CARLA simulator image, the actor has been spawned, and two vehicles crossed paths without any collision. The collision test report in the terminal also shows that no collision happened.





```
gabriel@gabriel-MS-7C94: /opt/carla-simulator/MT-scenario_runner-0.9.13
gabriel@gabriel-MS-7C94: ~/Desktop/carla-simulator
WARNING: Cannot spawn actor vehicle.nissan.patrol at position Location(x=106.999817, y=133.417389, z=1.000000)
WARNING: Cannot spawn actor vehicle.nissan.patrol at position Location(x=107.999817, y=132.417389, z=0.000000)
Here is the error degree:
degree: 49
x_degree: 1
y_degree: -1
z_degree: 0
ScenarioManager: Running scenario FollowVehicle

===== Results of Scenario: FollowVehicle ---- TIMEOUT =====

> Ego vehicles:
Actor(id=4671, type=vehicle.lincoln.mkz_2017);

> Other actors:
Actor(id=4677, type=vehicle.nissan.patrol);

> Simulation Information


|                                 |                     |
|---------------------------------|---------------------|
| Start Time                      | 2023-07-21 22:07:42 |
| End Time                        | 2023-07-21 22:08:42 |
| Duration (System Time)          | 60.06s              |
| Duration (Game Time)            | 60.03s              |
| Ratio (System Time / Game Time) | 0.999s              |



> Criteria Information


| Actor                      | Criterion            | Result  | Actual Value | Expected Value |
|----------------------------|----------------------|---------|--------------|----------------|
| lincoln.mkz_2017 (id=4671) | CollisionTest (Req.) | SUCCESS | 0            | 0              |
| nissan.patrol (id=4677)    | CollisionTest (Req.) | SUCCESS | 0            | 0              |
|                            | Timeout (Req.)       | FAILURE | 60.03        | 60             |
|                            | GLOBAL RESULT        | TIMEOUT |              |                |



=====
Not all scenario tests were successful
Destroying ego vehicle 4671
ERROR: failed to destroy actor 4671 : unable to destroy actor: not found
No more scenarios .... Exiting
WARNING: attempting to destroy an actor that is already dead: Actor 4671 (vehicle.lincoln.mkz_2017)
Iteration 1 finished.
(base) gabriel@gabriel-MS-7C94: /opt/carla-simulator/MT-scenario_runner-0.9.13$
```

I am using the latest scenario_runner-0.9.13 , and the scenario is FollowLeadingVehicle_1

Steps to Reproduce:

1. Load CARLA simulator and set up a scenario with an ego vehicle at (Location(x=107.999817, y=133, z=0.5)).
2. Attempt to spawn another actor at (Location(x=107.999817, y=132.417389, z=1.000000), Rotation(pitch=0.000000, yaw=49.000000, roll=0.000000)) using the spawn function.

3. Observe that the actor is spawned successfully and crosses paths with the ego vehicle without any collision, despite the expected collision.

Expected Behavior:

The `spawn` function should prevent the actor from being spawned at a location where it would collide with the ego vehicle. It should detect possible collisions and avoid such spawning.

Additional Information:

- I have consulted the CARLA documentation and API reference for information on the `spawn` function and collision detection.
- I have tried using different spawn locations and rotations, but the issue persists.
- I have also tested the scenario with other actor types, and the collision detection seems to be working correctly for them.

Please let me know if you need any further information or if there are any workarounds to address this issue. Thank you for your assistance in resolving this bug.



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

No branches or pull requests

1 participant

